THE NOVA RESEARCH CURRICULUM

Mathematics

Grade 7 · Term 1

Student edition

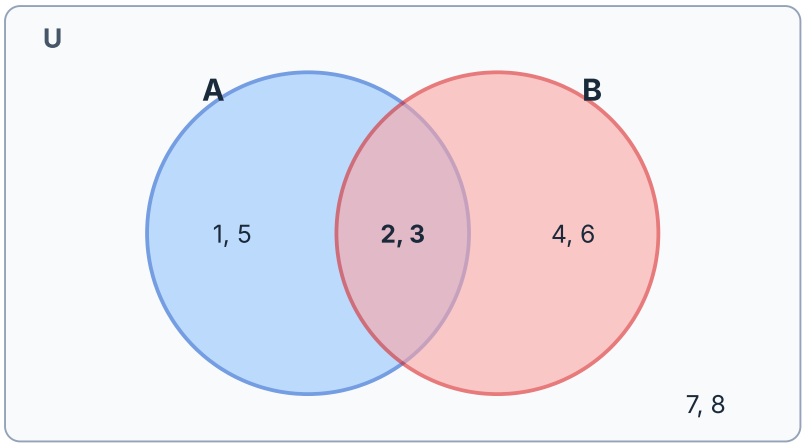
Outcomes: MA-7.1 rational numbers and sets; MA-7.2 proportion, profit and loss, tax, zakat and ushr.

Lesson notes

- **MA-7.1 Rational numbers (Weeks 1–4):** a rational number is p/q with q ≠ 0. Place on a number line; add, subtract, multiply and divide with signs; order rational numbers.

  ▶ **Try it on NovaSchool:** Zooming in on the number line — open **Simulations** on your dashboard.

- **MA-7.1 Sets (Weeks 5–8):** union (A ∪ B), intersection (A ∩ B), difference (A − B), complement (A′) within a universal set; Venn diagrams.



A = {1, 2, 3, 5}

B = {2, 3, 4, 6}

A ∩ B = {2, 3}

A ∪ B = {1, 2, 3, 4, 5, 6}

A − B = {1, 5}

A′ = {4, 6, 7, 8}

A Venn diagram: union, intersection, difference and complement.

- **MA-7.2 (Weeks 9–12):** direct proportion (more items, more cost) and inverse proportion (more workers, fewer days). Profit % = profit ÷ cost price × 100. Sales tax on a bill. Zakat: 2.5% of savings at or above the nisab, held for one lunar year. Ushr: 10% of produce on rain-watered land, 5% on irrigated land.
- **Watch out:** dividing by the selling price for profit %; mixing up direct and inverse proportion.

Workbook pages

Page	Task
1–2	Operations on rational numbers; order on a number line.
3–4	Set operations and Venn diagrams.
5–6	Proportion word problems; profit, loss and tax on a shop bill; zakat and ushr examples.

Question bank

#	Outcome	Question	Your answer
1	MA-7.1	Which is NOT a rational number?	3/4 / −5 / 0.25 / 5/0
2	MA-7.1	(−2/3) × (3/4) = ?	1/2 / −1/2 / −5/7 / 6/12
3	MA-7.1	A = {1, 2, 3}, B = {2, 3, 4}. A ∩ B = ?	{1, 2, 3, 4} / {2, 3} / {1} / {4}
4	MA-7.1	A = {1, 2, 3}, B = {2, 3, 4}. A ∪ B = ?	{1, 2, 3, 4} / {2, 3} / {1, 4} / {}
5	MA-7.1	A = {1, 2, 3}, B = {2, 3, 4}. A − B = ?	{1} / {4} / {2, 3} / {1, 4}
6	MA-7.2	4 pens cost Rs 120. What do 10 pens cost?	Rs 240 / Rs 300 / Rs 400 / Rs 480
7	MA-7.2	6 workers take 10 days. How long do 12 workers take?	20 days / 5 days / 10 days / 3 days
8	MA-7.2	Cost Rs 500, sold for Rs 600. Profit %?	16.7% / 20% / 100% / 10%
9	MA-7.2	Zakat on savings of Rs 200,000 held for a year (above nisab)?	Rs 2,000 / Rs 5,000 / Rs 20,000 / Rs 25,000
10	MA-7.2	Ushr on 1,000 kg of wheat from rain-watered land?	100 kg / 50 kg / 25 kg / 10 kg